terminal and the holes in the *p*-side are attracted by the negative terminal. This increases the immobile ion concentration at the junction. The net effect is the widening of the depletion region leading to an increase in the barrier potential. Consequently, the majority charge carriers from both sides experience a great potential barrier to cross the junction. This reduces the diffusion current across the junction drastically.

Yet, a small current flows across the junction due to the minority charge carriers in both regions. The reverse bias for majority charge carriers serves as the forward bias for minority charge carriers. The current that flows under a reverse bias is called the reverse saturation current**.** It is represented as I_s.

The reverse saturation current is independent of the applied voltage and it depends only on the concentration of the thermally generated minority charge carriers. Even a small voltage is sufficient enough to drive the minority charge carriers across the junction.

Note

The reverse saturation current of a silicon diode doubles for every 10 °C rise in temperature.

10.3.3 Characteristics of a junction diode

i) Forward characteristics

It is the study of the variation in current through the diode with respect to the applied voltage across the diode when it is forward biased.

The *p-n* junction diode is forward biased as shown in Figure 10.14(a). An external resistance (R) is used to limit the flow of current through the diode. The voltage across the diode is varied by varying the biasing voltage across the DC power supply. The forward bias voltage and the corresponding forward bias current are noted. A graph is plotted by taking the forward bias voltage (V_F) along the x-axis and the current (I_F) through the diode along the y-axis. This graph is called the **forward V-I characteristics** of the *p-n* junction diode and is shown in Figure 10.14(b). Four inferences can be brought out from the graph:

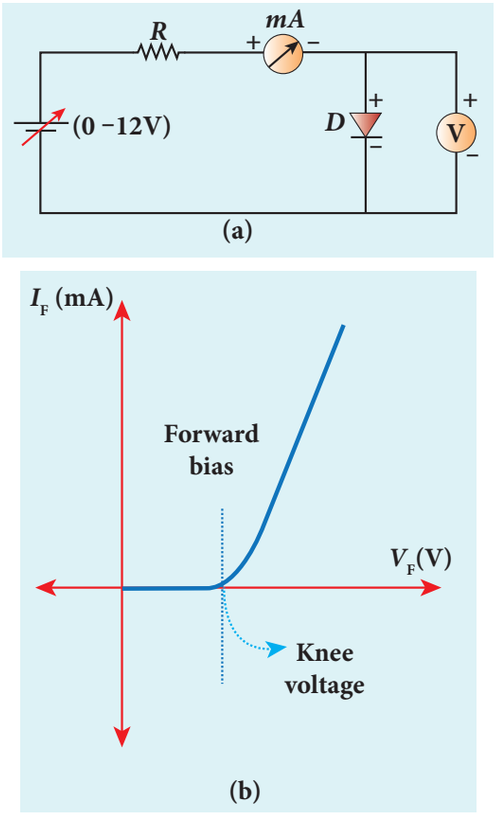


Figure 10.14 *p-n* junction diode
(a) Diode under forward bias
(b) Forward characteristics

(i) At room temperature, a potential difference equal to the barrier potential is required before a reasonable forward current starts flowing across the diode. This voltage is known as **threshold voltage or cut-in voltage or knee voltage** (V_{knee}). It

UNIT-10_XII-Physics_Vol-2_New.indd 205 21-07-2020 15:35:15